Explainability as a non-functional requirement in an AI-assisted veterinary clinical system: a case study

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Abstract

Context. AI-assisted diagnostic modules in veterinary clinical-management systems frequently lack explainability operationalized as a verifiable requirement.
Objective. To determine which explainability requirements, operationalized as verifiable non-functional requirements (NFRs), are necessary and sufficient for the different user profiles (technical and non-technical) of an AI-assisted veterinary clinical system.
Method. Applied single-case study grounded in Requirements Engineering. Requirements were elicited through semi-structured interviews and direct observation, analyzed qualitatively, operationalized as functional and non-functional requirements, and linked through requirements traceability.

Results. Open coding of 16 semi-structured interviews yielded 167 open codes, consolidated into 47 axial codes across 7 categories; closed-coding saturation was reached (1.33 new axial codes on average in the final three interviews, 2.84% of the cumulative total, below the 5% threshold). Acceptance of AI-assisted diagnostic suggestions was consistently conditional: participants required the veterinarian's ability to review and correct suggestions (6/16) and grounding in validated literature and data-protection safeguards (6/16), with general moderate-to-high acceptance (11/16) and AI treated as decision support rather than a substitute for professional judgement. These conditions were operationalized as non-functional requirement RNF-18, traced to the AI-assisted diagnostic use case CU-06.
Conclusions. Explainability for AI-assisted veterinary diagnostic support is best operationalized around contestability and provenance of suggestions rather than technical model interpretability alone, and can be linked, through requirements traceability, to concrete verifiable requirements. Broader validation across additional cases and a second independent coding round remain as future work.

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Keywords

Explainability; Non-functional requirements; AI-assisted diagnosis; Veterinary informatics; User studies

Declarations

Funding

The authors received no specific funding for this work.

Competing Interests

The authors declare no competing interests.

Ethics approval and consent to participate

Participation in the requirements-elicitation activities was voluntary and based on written informed consent. Participant information is handled using anonymized identifiers, and identifying materials are maintained separately in the restricted evidence area in accordance with the project's data-protection procedures and Ecuador's *Ley Orgánica de Protección de Datos Personales* (LOPDP).

Pending confirmation before submission: the final manuscript must state whether formal approval was granted by an institutional ethics committee or equivalent body and, if applicable, provide the official approval identifier. No approval number is reported here because it has not yet been confirmed. Before asking the course instructor directly, check

`08_Etica/Aval_Institucional.pdf` in the project repository (referenced as a condition of admission, gatekeeper G8, in the Entrega 4 rubric) – the identifier, if one was issued, is most likely documented there.

Consent for publication

Pending verification before submission. The final statement must reflect the exact scope of the signed consent documents regarding the use and publication of anonymized participant information.

Data, Materials and/or Code availability

The research materials supporting the requirements-engineering phase include anonymized interview transcripts, thematic-coding materials, requirements artefacts, traceability information, and related project documentation. Potentially identifying materials, including original consent documents and multimedia evidence, are maintained separately in the restricted evidence area and are not intended for public dissemination.

The anonymized replication package is deposited on Zenodo: https://doi.org/10.5281/zenodo.22238486. *Pending verification before submission: this DOI currently points to the automatically generated deposit under an MIT license; it must be replaced with the DOI of the curated CC BY 4.0 dataset package once that deposit is finalized (see `07_Datos/registro_deposito.md`).* The protocol for the Enfoque 3 empirical component is registered on OSF: https://osf.io/r5p8d/ (registered 1 August 2026).

Author contributions

Partial draft based on verified repository artefacts; must be reviewed and completed by all authors before submission, preferably following the CRediT taxonomy.

Marcillo Ponce, A.J.: thematic coding (open coding of interviews P09–P16; axial consolidation), thematic-saturation analysis, traceability-matrix verification, manuscript writing (original draft and revisions), project administration for this submission. **Amagua Sacón, R.W.**: traceability-matrix corrections and verification, open-coding saturation analysis script and figure. **Vera Gómez, A.A.**: Enfoque 3 empirical-component design, OSF protocol registration, validation-instrument design, confirmation of protocol deviations. **Mesías Quijije, J.A.**: *pending – to be completed by the author.* **Barrionuevo Fuentes, C.D.**: *pending – to be completed by the author.* **Guerrero Ulloa, G.C.**: academic supervision.

All authors read and approved the final manuscript.

Use of AI-assisted technologies

During preparation and revision of this manuscript, an AI-assisted large language model (Claude, Anthropic) was used interactively, with human oversight, exclusively as a support tool for verification and language improvement. Its use was limited to: (i) checking the consistency, clarity, grammar, spelling, and wording of text previously prepared by the authors; (ii) verifying bibliographic and other factual data against the information and sources used by the authors; and (iii) suggesting improvements to the writing and overall clarity of the manuscript. The AI tool was not used to generate original or substantive content, including research data, thematic codes, categories, analysis, results, methodology, interpretations, conclusions, tables, figures, or research decisions. All suggestions were reviewed by the authors before inclusion. The authors remain fully responsible for the accuracy, provenance, and scientific integrity of all content and for the final version of the manuscript.

Acknowledgements

The authors thank the veterinary professionals, administrative personnel, and other participants who contributed to the requirements-elicitation and validation activities of the SGCV-IA project. Participant identities are not disclosed in this manuscript in accordance with the project's anonymization and data-protection procedures.